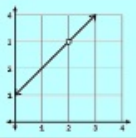
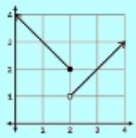
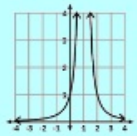
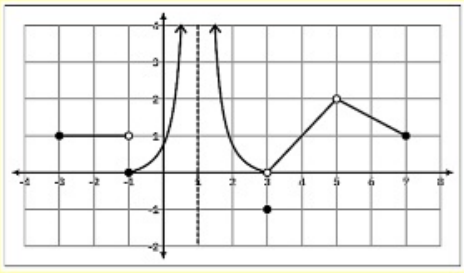
LIM-2.A.1 Types of discontinuities include removable discontinuities, jump discontinuities, and discontinuities due to vertical asymptotes.

1-10 Types of Discontinuities

There are two types of discontinuities: removable and nonremovable.

Category	Name (Type)	Example Equation	Graph	What happens to the function at the point of discontinuity?
Removable	**Removable** (hole)	$f(x) = \frac{x^2-x-2}{x-2}$ $\frac{(x-2)(x+1)}{x-2}$		At x = 2 • limit exists • value of the function is not equal to the limit
Nonremovable	**Jump**	$f(x) = \begin{cases} 4-x & x \le 2 \\ x-1 & x > 2 \end{cases}$		At x = 2 • left- and the right-hand limits exist but they are not equal • value of the function can equal either or neither limit
	Infinite (vertical asymptote)	$f(x) = \frac{1}{(x-1)^2}$		At x = 1 • left- or right-hand limits (or both) equal $\pm\infty$ • value of the function may or may not exist

Example 1: The graph at the right is $f(x)$. Determine all x values where $f(x)$ is not continuous and classify each point.



$x=-1$ jump

$x=1$ infinite

$x=3$ removable

$x=5$ removable

Example 2: $g(x) = \frac{x^2+3x-40}{x^2-3x-10}$. List & classify the x values where $g(x)$ is not continuous.

$$g(x) = \frac{(x+8)\cancel{(x-5)}}{(x+2)\cancel{(x-5)}} = \frac{x+8}{x+2}$$

hole
$x-5=0$
$x=5$

VA: $x+2=0$
$x=-2$

$x=5$ removable $\quad x=-2$ infinite

What would make a function continuous at a point?

A function is continuous if the limit of the function at a point equals the value of the function at that point.

$$\lim_{x \to a} f(x) = f(a)$$

Example 3: Consider the piecewise defined function for $h(x)$ below. Determine all x-value(s) where $h(x)$ is discontinuous and classify each discontinuity.

$$h(x) = \begin{cases} 2x-1 & x < -2 \\ x^2-9 & -2 \le x < 1 \\ 3x+2 & 1 \le x < 3 \\ 4x-1 & x > 3 \end{cases}$$

$x=-2$ $\lim_{x\to -2^-} h(x) = \lim_{x\to -2^-}(2x-1) = -4-1 = -5$

cont. $\lim_{x\to -2^+} h(x) = \lim_{x\to -2^+}(x^2-9) = 4-9 = -5$

$\lim_{x\to -2} h(x) = -5$ $h(-2) = 4-9 = -5$

$\boxed{x=1 \text{ jump}}$ $\lim_{x\to 1^-} h(x) = \lim_{x\to 1^-}(x^2-9) = 1-9 = -8$

$\lim_{x\to 1^+} h(x) = \lim_{x\to 1^+}(3x+2) = 3+5 = 5$ $\neq$

$\lim_{x\to 1} h(x) = \text{DNE}$

$\boxed{x=3 \text{ removable}}$ $\lim_{x\to 3^-} h(x) = \lim_{x\to 3^-}(3x+2) = 9+2 = 11$

$\lim_{x\to 3^+} h(x) = \lim_{x\to 3^+}(4x-1) = 12-1 = 11$

$\underbrace{\lim_{x\to 3} h(x) = 11 \quad h(3) = \text{undef.}}_{\text{hole}}$

Example 4: The piecewise function for $f(x)$ is below. Is $f(x)$ continuous at $x = 2$? Justify your answer.

$$f(x) = \begin{cases} x^2 + x - 3 & x < 2 \\ 5x - 7 & x \geq 2 \end{cases}$$

$f(2) = 5(2) - 7 = 3$

$\lim_{x \to 2} f(x) = 3$

$\lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} (x^2 + x - 3) = 4 + 2 - 3 = 3$

$\lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} (5x - 7) = 10 - 7 = 3$

$f(x)$ is cont. at $x = 2$ b/c $\lim_{x \to 2} f(x) = f(2)$

Example 5: Find the value for c that makes the piecewise function $q(x)$ continuous at $x = 1$.

$$q(x) = \begin{cases} x - 3 & x \leq 1 \\ x^2 + cx - 5 & x > 1 \end{cases}$$

$q(x)$ is cont. at $x=1$ $\Longrightarrow$ $\lim_{x \to 1} q(x) = q(1)$

$$q(1) = \lim_{x \to 1^-} q(x) = \lim_{x \to 1^-} (x-3) = 1-3 = -2$$

$$\lim_{x \to 1^+} q(x) = \lim_{x \to 1^+} (x^2+cx-5) = 1+c-5 = c-4$$

$$\lim_{x \to 1^-} q(x) = \lim_{x \to 1^+} q(x)$$

$$-2 = c-4$$

$$\boxed{2 = c}$$